

The Free-viewing VR Presentation Tool

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ABSTRACT

Introducing a user-friendly Free-viewing VR Presentation Tool in Unity for seamless VR experiment design and execution. It's suitable for users with limited coding skills, supports eye-tracking integration, external recording, and suits various applications like psychology, game design, and marketing. Its simplicity, versatility, and potential for future improvements make it a valuable asset for efficient VR experiment design.

Index Terms: Virtual Reality (VR)—presentation tool—psychology—free-viewing user studies;

1 INTRODUCTION

VR hardware offers an excellent stimuli presentation platform because of its immersive potential and incomparable control over the user's environment, allowing precise adjustment of every aspect of the visual field, e.g. ambient light or peripheral cues. This is especially important while gathering data from the user's gaze or physiological reactions, as they are very sensitive to even the smallest changes in the surroundings. Therefore, the popularity of VR as a research platform is growing exponentially, both in the fields directly related to VR environments - e.g. designing the virtual agents or testing behaviors of virtual crowds, but also in the broader fields of experimental psychology, neuroscience, user experience research, or marketing. Nevertheless, none of the popular general-purpose stimuli presentation software, both commercial such as Presentation by Neurobehavioral Systems (<https://www.neurobs.com/>), and open source such as PsychoPy [4], currently supports the preparation of the studies in VR environment. The existing tools, such as Biomotion Lab Toolkit for Unity Experiments (bmlTUX; [1]), Unity Experiment Framework (UXF; [2]), Unified Suite for Experiments (USE; [5]) or Virtual Reality Scientific Toolkit (VRST; [3]), facilitate creating experimental designs in Unity or even support streaming psychophysiological data but do not directly tackle the basic procedures known from vision research in which participants are asked to "just look" at the presented stimuli (i.a. "free-viewing procedure"). In these tools, researchers are compelled to either create specific

scenes for participant presentation or write code/commands, often requiring expertise in Unity scene creation and coding, or custom-made coding with support from 3D software developers, thereby posing time and cost barriers that impede the wider adoption of VR hardware in the research area.

The need for delivering simplified presentation software is indeed high, as evidenced by posts on software discussion groups, for example, the PsychoPy board <https://discourse.psychopy.org/>. Such software needs to support the creation and implementation of the test routines, as well as data acquisition, recording, and storage. In particular, they need to be able to present stimuli for a predefined time, log user responses, and send triggers to external measurement devices like EEG or physiological measurement equipment, which is needed to synchronize the data from different channels. Importantly, it cannot require operating directly in the real-time development platforms (e.g. Unreal Engine or Unity), but rather should offer a user-friendly interface with a low entry threshold for people with no previous experience in programming in the virtual environment, or even programming at all.

Here we describe a solution developed in Unity designed to offer functionality equivalent to the aforementioned general-purpose stimuli presentation software.

2 OVERVIEW

Our Free-viewing VR Presentation Tool enables researchers to design and conduct an experiment in the VR environment by specifying its content and time course with nothing more than a collection of one-line commands assembled in a configuration file in .csv format, that is loaded and initialized using Inspector GUI - the front end of the Unity pre-defined scenario.

Additionally, our solution enables integration with an eye-tracking device developed for HTC Vive pro goggles (in our case Pupil Labs), as well as implements TCP/IP communication functionality with external recording modules (in our case LabView) by sending triggers (signaling the occurrence of a given stimulus type) to a predefined internet address. Therefore, in addition to gaze tracking and pupil dilation measurements using Pupil Labs HTC Vive Pro more sophisticated features and analysis can be accessed using different external recording softwares with different sensors. To date, we have successfully tested the solution provided by Plux Biosignals Bitalino in Researcher Kit with the following sensors: Electrodermal Activity (EDA); Electrocardiography (ECG); light (lux) sensor. However, there is a potential to implement also other sensors, including electromyography (EMG), electroencephalography (EEG), respiration, temperature, force, or blood oxygen saturation. It is worth noticing that while external software components need to be

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used the data synchronization is achieved with remarkable precision, thanks to the triggers sent in real time.

Significantly, our solution does not require virtually any experience with the Unity environment. Before the research session, the stimuli sequence scenario gets loaded within Unity to the prepared scriptable object from researcher-provided .csv via Inspector GUI of the Scenario Manager component that allows for convenient session necessities setup. Each scenario comprises also record rows containing unique stimuli trigger IDs with their image or video source paths, duration, scale, and background color.

The Scenario Manager supports all the most common picture (.jpg, .png) and video (.asf, .avi, .mov, .mp4) formats. The configuration file can also control the flow of the experimental session via researcher-predefined functional trigger IDs for managing eye-tracking recording and calibration, and a special set of commands that control the starting, stopping, and pausing of the experiment. When connected with eye-tracking recording software, GUI also allows initializing the on-demand calibration procedure and repeating it as many times as possible to obtain the expected accuracy, before the experiment start or during the pause.

The configuration file is stored in .csv format, therefore it could be edited manually in a spreadsheet or text editor for a simple design, but also generated by an external script and then stored and passed to the tool. That way the more complex research designs, requiring grouping stimuli into blocks or stimuli randomization could be managed using individual configuration files, operating on the same set of stimuli and loaded immediately using the single GUI component.

The data acquisition is made with the use of Pupil Labs HMD Eyes Unity package components managed by custom scripts developed for the project and with concurrent use of Pupil Labs Pupil Capture software and Pupil Labs HTC Vive Pro Binocular Add-on. During the research session parts that have recording turned on the software sends the just-loaded trigger ID to the specified IPv4 address at the specified port, as well as to the Pupil Capture in the form of an annotation. After each recording, the stored eye-tracker data are ready for loading in Pupil Labs Pupil Player software and from there, it is possible to export the data of the desired researcher-predefined parameter types into the human-readable .txt files.

The tool allows for various customizations via the Inspector GUI of the Scenario Manager component, such as changing the target host IP and port, changing the functional triggers IDs, defining the background color and the name of the manual calibration background image that is to be automatically loaded from within the researcher-defined scenario resources folder. Disabling/enabling showing background images during calibration and keeping them hidden or showing them after manual calibration is also configurable. It is also possible to preview the loaded .csv scenario record rows from within the Inspector GUI of the scriptable object in which the loaded scenario data gets stored.

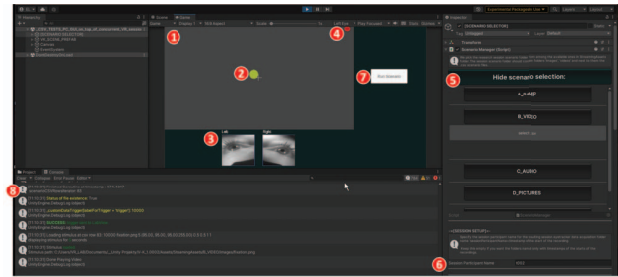


Figure 1: The experimenters' view. 1. The screen as seen by the participant - in this example, the participant sees the fixation cross. 2. The green dot shows the real-time participants' gaze position, as tracked by the eye-tracking device - in this example Pupil Labs. 3. Preview of participant's eyes from an eye-tracking camera. 4. Indication of data recording. Recording starts and stops automatically as defined by the scenario or could be started and stopped manually using "R" button on the keyboard. 5. Scenario selection GUI. Each button is associated with a directory with configuration file/s. The number, as well as names of scenarios, could be changed in the GUI. 6. Participant name used as a unique ID for all recorded data. 7. GUI element for initializing selected scenario. 8. Console for displaying the log information, in particular scenario loading, stimuli loading and trigger sending to external measuring equipment - in this example LabView.

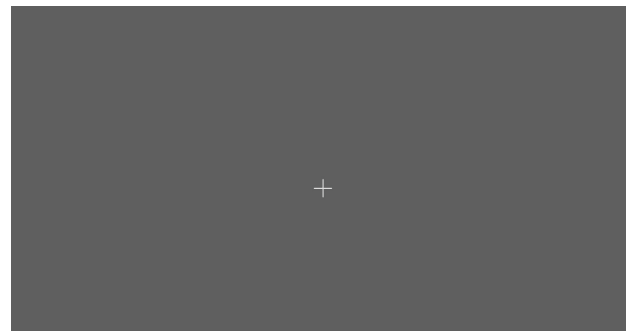


Figure 2: The participant's view (the pre-stimulus fixation cross)

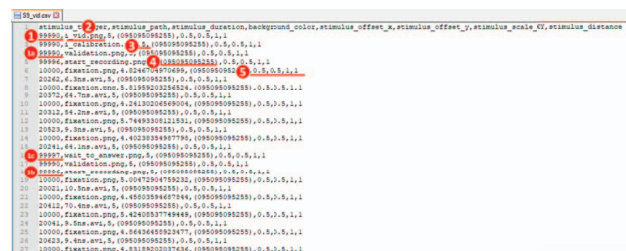


Figure 3: The exemplary configuration file. 1. Stimulus trigger sent to external recording software. The special triggers are defined in the GUI and include 1a. Time limitless trigger means that the stimulus is displayed until the press of a defined key. 1b Start recording 1c Stop recording. Triggers require a 5-digit format. 2. Stimulus path - in this case - filename. 3. Stimulus duration. In the case of the time limitless trigger, this value is ignored. 4. Background definition in RGBA colour model. 5. Definition of stimulus onset, offset, scale, and distance in 3D space. For a free-viewing experiment, those factors were set as default.

3 APPLICATION

Importantly, the versatile nature of this software allows for various applications in fields such as psychology and neuroscience, but also game design and user experience, or even marketing and advertising. Virtually any material could be displayed, including but not limited to pictures, movies, advertisements, in-store designs, gameplays, virtual agents, or virtual crowds.

One key application is the adaptation of experimental designs used in these disciplines to be presented within a VR environment. This adaptation provides a superior level of control over the external environment, enhancing the precision and reliability of experimental conditions.

Another notable advantage is the software's suitability for quick pre-tests, particularly for evaluating multiple designs. The flexibility of the software allows for the addition or modification of conditions without the need for extensive preprogramming, streamlining the research process.

Additionally, the software finds applicability in studies where monitoring user responses in real time is crucial. This could be used in studies involving virtual agents, where the observation of physiological responses might uncover implicit reactions to virtual stimuli. It also proves beneficial for studying gaze patterns in virtual crowds, offering insights into user behavior within simulated environments. In summary, the software's multifaceted capabilities make it a valuable tool for diverse research scenarios, providing researchers with a range of options to enhance their studies.

4 SUMMARY

Free-viewing VR presentation tool provides a template for conducting studies using the free-viewing procedure. Thanks to this, it enables rapid experiment design by simply specifying file paths and defining the start, end, and recording pauses. Therefore, even a researcher with no previous experience with the Unity environment and little or no coding knowledge is able to run a simple free-viewing experiment on his own.

The Inspector GUI has a very low entry threshold, working on similar principles to the tools commonly used in experimental psychology, such as Psychopy [4]. We pilot-tested the GUI on a small group of undergraduate and graduate students, showing that it took less than an hour to teach the research assistant to operate the VR presentation tool on the GUI level. Despite being targeted at novice coders, also advanced coders will benefit from using the free-viewing VR presentation tool, as they will not need to focus on coding the Unity scenes while still being able to implement any free-viewing based research designs and stimuli combinations via the customizable configuration files, that could be created and easily exported from any common programming environment.

The current limitations of the tool are partially caused by the limitations of the convenience and the performance of the currently available necessary hardware and partially by the hardware used for which the tool was developed. Until now, tools like the introduced one will remain for quite narrow hardware use cases, which in a matter of the described tool are HTC Vive, Vive PRO, or Vive Cosmos, Pupil Labs HTC Vive Pro Binocular Add-on and Pupil Labs HMD Eyes. Therefore, at this point, our research tool is not yet widely accessible, but in the future it might become a shareable and common-use type of tool.

Given the possibilities of the developed tool, its' next iterations will explore the limits of precision, efficiency, and holistic data acquisition and processing. The other possible development direction is creating a library of pre-defined scenes based on the same architecture, but allowing more diverse user interactions with the environment.

5 ACKNOWLEDGMENTS

This work was supported by the National Science Centre in Poland (grant number 2017/25/B/HS6/00758). Work of R. Sterna was financed from the budget for science in the years 2019-2023, as a research project "The impact of virtually generated characters: human aspects of bots" (project number: DI2018 015848) under the Diamond Grant programme granted by the Ministry of Science and Higher Education of Poland. Work of M. Igras-Cybulska was supported by program Lider grant nr 0230/L-11/2019 by the National Center for Research and Development, Poland. The research has been supported by a Visibility and Mobility module and grants supporting publications in top journals and publishers under the Strategic Programme Excellence Initiative at the Jagiellonian University.

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